

1. How many times do you have to roll a fair die for the expected number of sixes that result to be 1? Give a proof.
2. A frog sits on one of a row of lily pads that extends infinitely far to the left and the right. Every minute, there is a $1/2$ probability that he jumps one pad to the left, a $1/3$ probability that he jumps one pad to the right, and a $1/6$ probability that he stays where he is. What is the expected lily pad for him to be on after an hour? *Hint:* Linearity of expectation.
3. We know that, if X and Y are independent random variables, then $E(X)E(Y) = E(XY)$. If we just know that X and Y are random variables such that $E(X)E(Y) = E(XY)$, can we conclude that X and Y are independent?
4. *Markov's inequality* states that, if X is a non-negative real-valued random variable and a is a real number greater than zero,

$$P(X \geq a) \leq \frac{E(X)}{a}.$$

Let I_a be the random variable that is 0 if $X < a$ and 1 if $X \geq a$. Show that $aI_a \leq X$ and hence prove Markov's inequality.